Text Generation using TF and Keras

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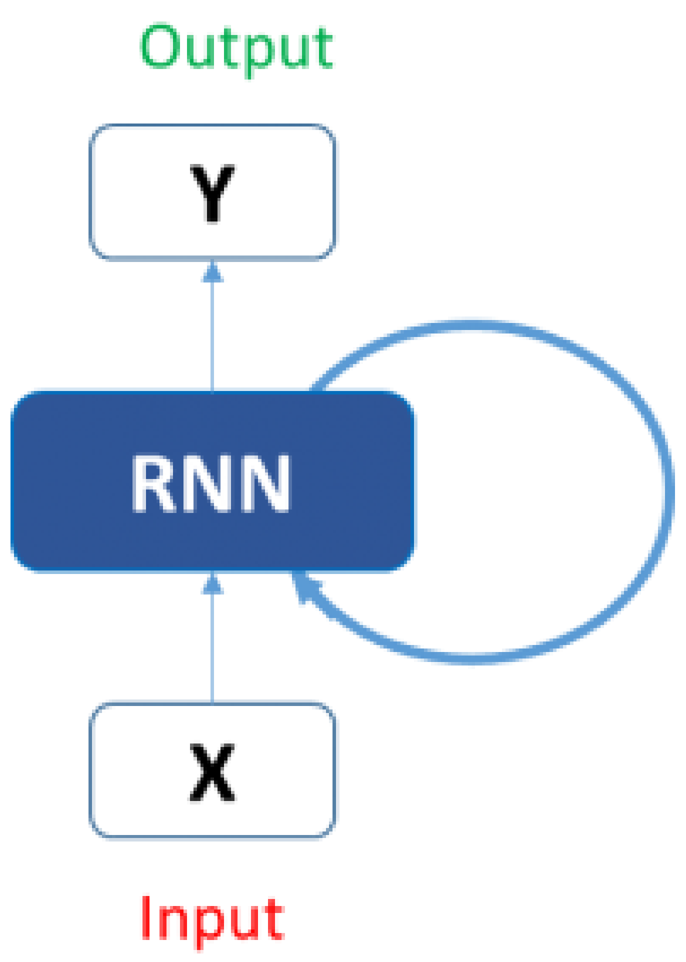
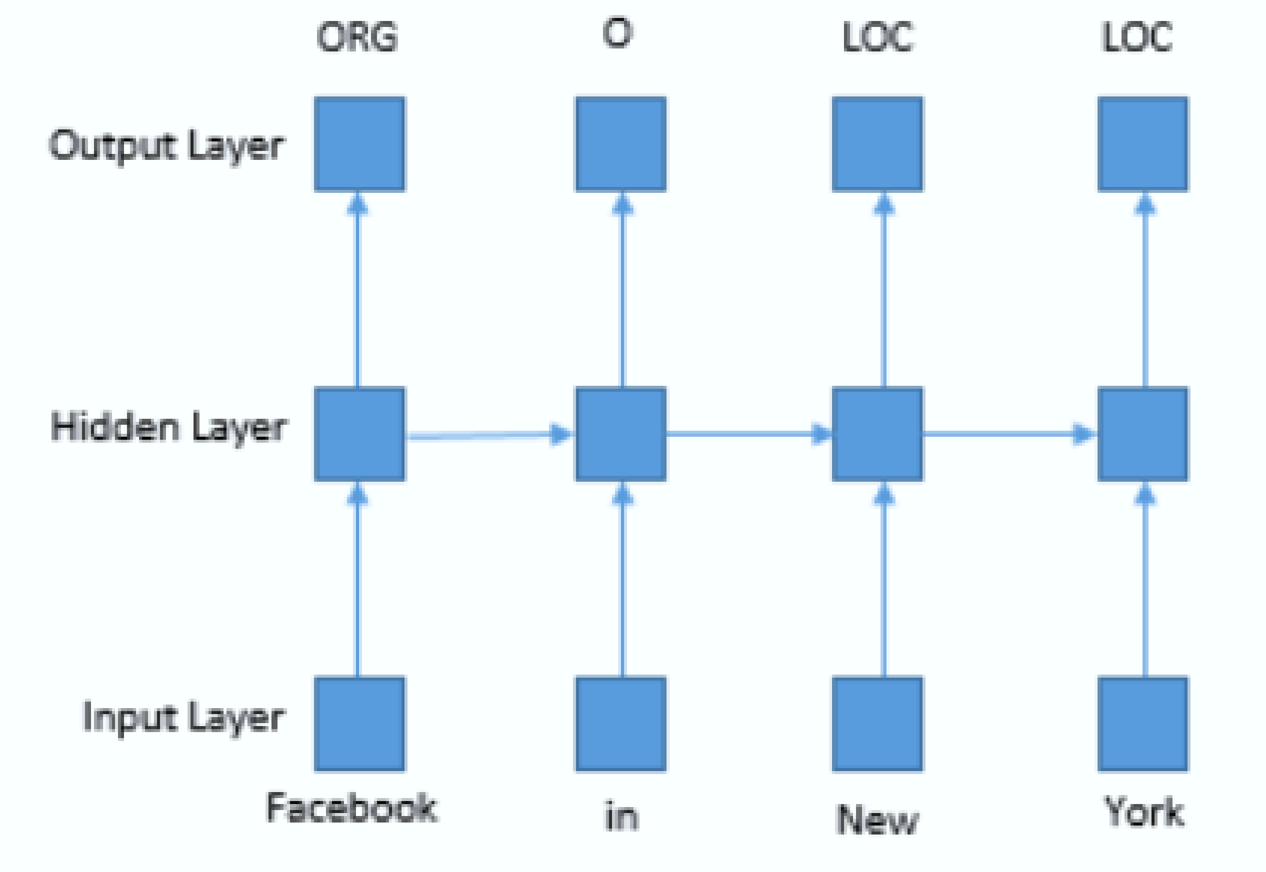
RNN

- Recurrent neural networks (RNNs) are one of the states of the art algorithm in deep learning especially good for sequential data (like text data, time series, financial, weather forecasting, etc.)
- RNN is considered the perfect NN model for used in NLP as it's “recurring” aspect of the model provides a better result in text generation



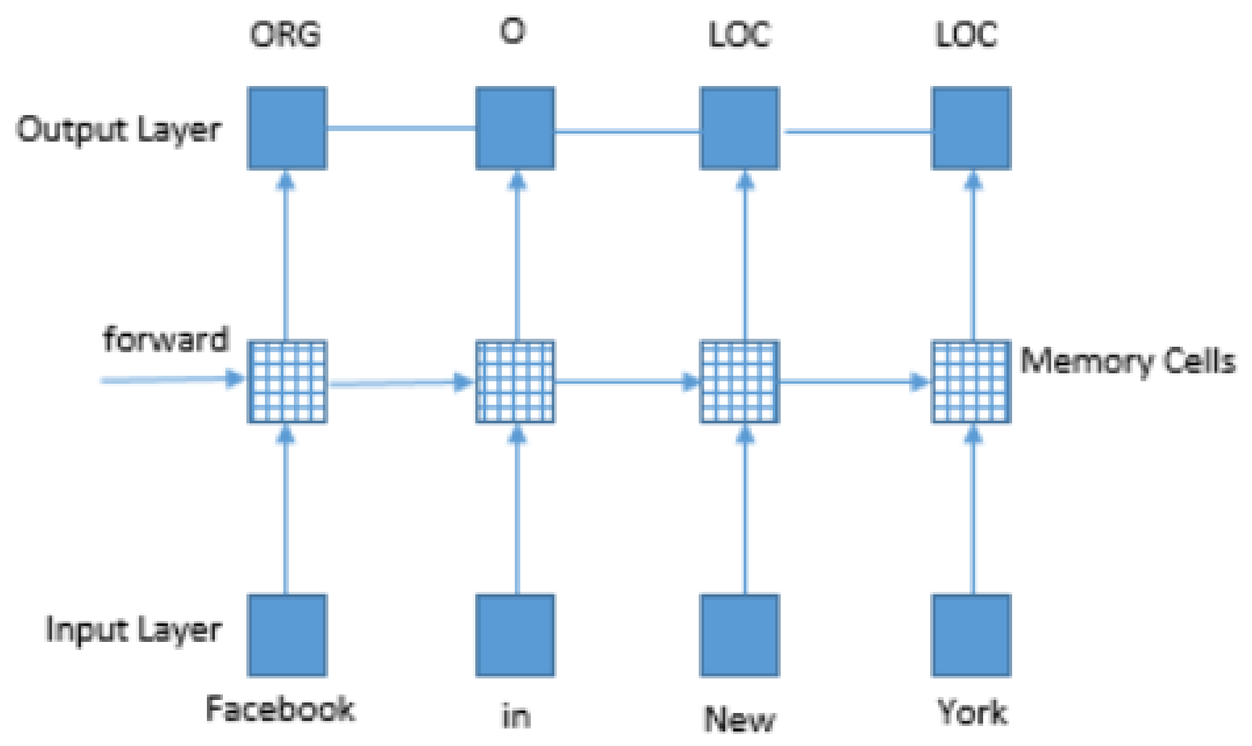
RNN in NLP

- RNS are particularly used if the prediction has to be at word-level for instance Named-Entity-Recognition (NER) or Part of Speech (POS) tagging.
- It stores the information for current features as well as neighboring features for prediction.
- Its memory based on history information enables the model to predict long distance features



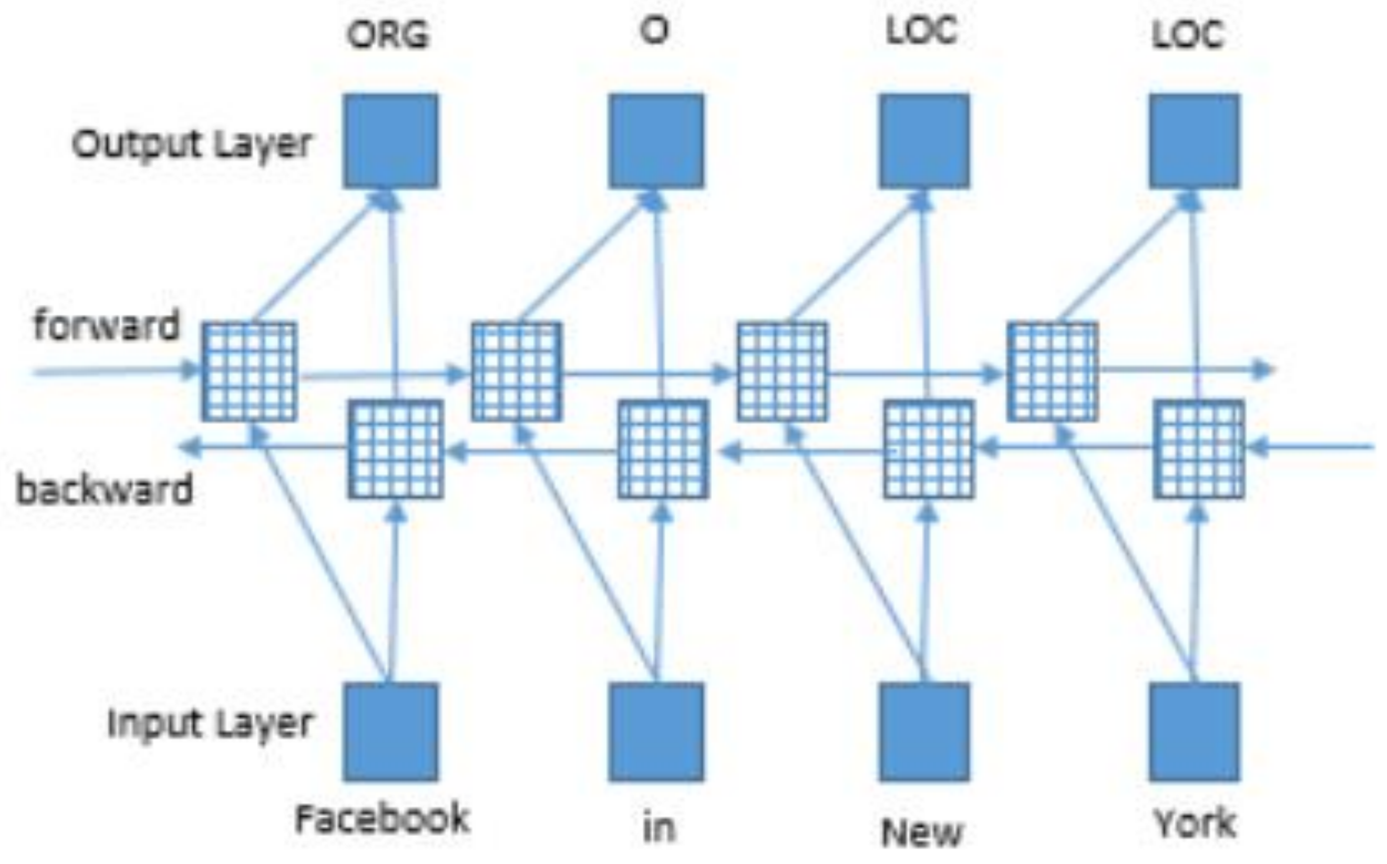
Use of LSTM in NLP

- Due to RNN only being able to learn about recent events, LSTMS are used to understand context.
- Using LSTM, a hidden layer is used called the memory cells where they can store information for longer usages so long-term information can continuously be factored.



Bidirectional LSTM

- Bidirectional LSTM has access to both past and future input features for a given time
- This is highly recommended to use in NLP due to there being sequence tagging or new named entities used as inputs



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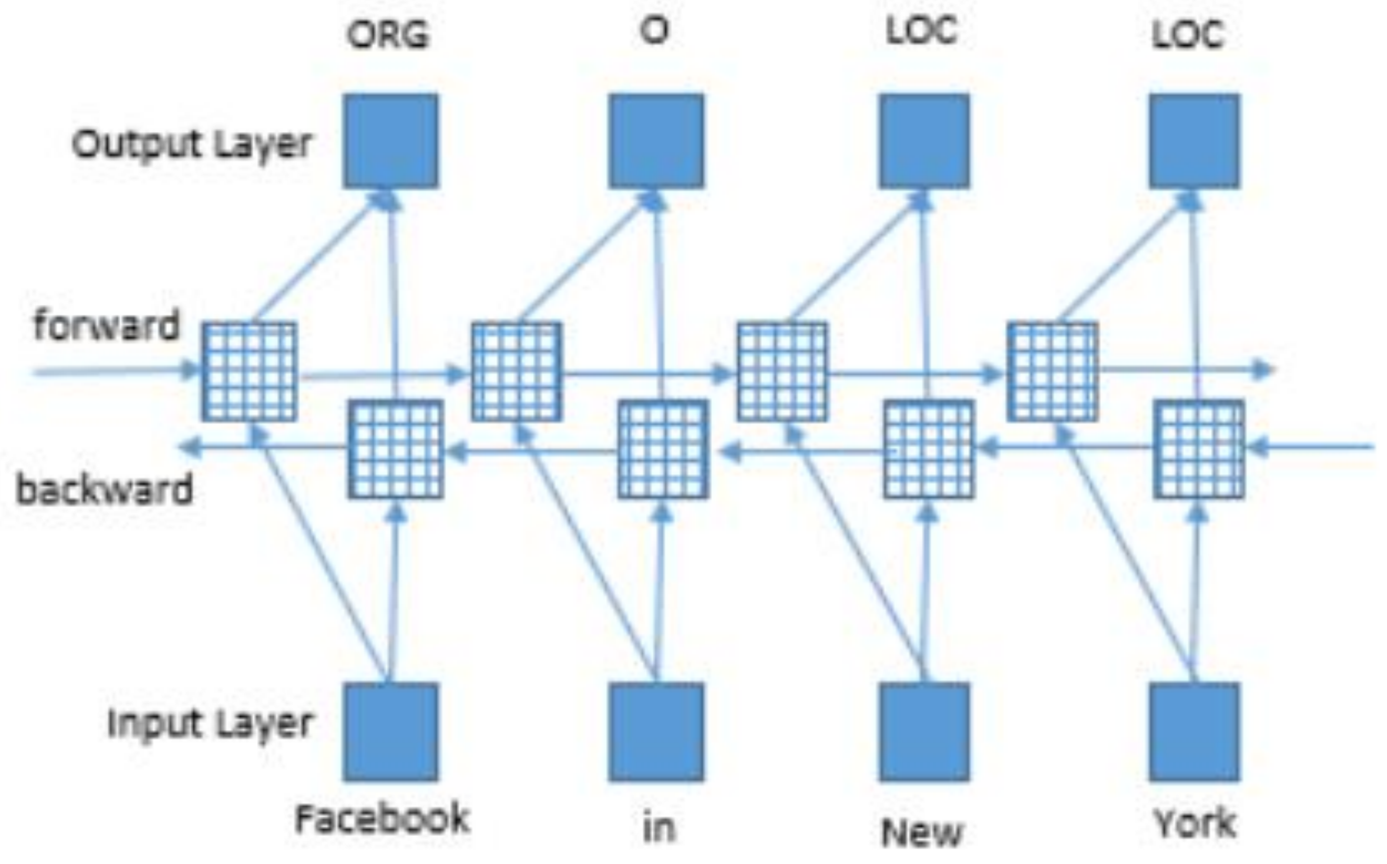

Standard Vocabulary Size

- The standard vocabulary size used for text generation is 256 bits based on the UTF-8 encoded sequence
- UTF-8 in this case is used due to the text generation sequence in this model does not include emoji's or anything else.
- Newer emoji models or characters that is uncommon to the English literature such as Japanese kanji (龙) or Greek symbols (λ) use the BBPE model (Byte-level BPE) where up to 130 thousand characters are provided.

Standard Vocabulary Size

Newer Unicode characters such as emoji's use the BPE model which looks like the example below'

Say we have an example text such as "That's Great 👍", the conversion to BPE and further on to the UTF-8 will look like

T	54
h	68
a	61
t	74
’	2019
s	73
	20
g	67
r	72
e	65
a	61
t	74
	20
👍	1F44D

BBPE Model

T	54
h	68
a	61
t	74
�	e2
�	80
�	99
s	73
	20
g	67
r	72
e	65
a	61
t	74
	20
�	f0
�	9f
�	91
�	8d

UTF-8 Sequence